

When is NoSQL the Best Choice for Database Management?

NoSQL databases can handle unstructured data. Find out how they differ from other databases, and when and where they should be used.



Different types of databases can be used for managing structured and unstructured data. Relational databases are used for structured data, which is formatted and has well-known types. NoSQL databases are used for unstructured data. A database management system typically has an interface, a query processing engine, metadata processor, indexing engine, and storage manager. These components help in querying and storing data into the DBMS. Earlier, blobs (an acronym for binary large objects) were used for storing

unstructured data. Tabular data stores evolved to columnar data stores to handle data that is not normalised.

NoSQL databases differ from an SQL DBMS in the way data is available — unstructured or structured. SQL is the query language for relational databases. These have well defined schema. Relational databases are used in transactional or reporting scenarios. NoSQL databases are used to handle unstructured databases in data lake scenarios. The data can be document based and key value type, or can be graph based. Fields can be added

dynamically, just like in a document.

The most popular NoSQL databases are MongoDB, Apache CouchDB, Apache HBase, Oracle NoSQL, Apache Cassandra, Neo4j, BigTable, Riak, Objectivity Infinite Graph, and MariaDB. Many enterprises are choosing NoSQL databases to handle data from multi-channel and IoT apps.

Data types

NoSQL database management systems were developed to handle key value data types, wide column data types, documents, and graph data types. A key-value data store is based on dictionary/hash tables. Document data stores can handle data formats like XML, YAML, and JSON. Graph data stores can handle graph based data. NoSQL databases are performant and scalable. REST, Graph QL, and other NoSQL query methods are used for accessing data from NoSQL databases.

SQL DBMSs handle data types like int, char, varchar, binary, varbinary, float, decimal, double, bit, blob, bool, longtext, tinytext, mediumtext, and text. Relational data format is used for transactional data processing like commerce ordering, accounting, and payment systems.

Programming language support

A structured query language is used for creation, updation, modification, and deletion of schema and data. In the design phase, a relational database (SQL DBMS) is designed for its schema and database procedures. NoSQL DBMS is based on a dynamic schema for handling unstructured and semi-structured data. Columns can be